

(PART A – 2 Marks)

1. What are the various sizes supported by buttons in Bootstrap?

- btn-lg
- btn-md
- btn-sm
- btn-block

2. Specify the types of container class in Bootstrap with their purpose.

- container – Creates a responsive fixed-width container that adapts at different breakpoints.
- .container-fluid – Provides a full-width container that spans the entire viewport.
- .container-{breakpoint} – Acts as a full-width container until the specified breakpoint, then becomes fixed-width.

3. Give the main features of Figma used in UI design.

- Cloud based
- Real-time collaboration
- Cross platform
- Plugins and Widgets

4. Mention the key components of Bootstrap.

- alerts
- navbar
- buttons
- progress bar

5. Demonstrate a 3-column responsive layout that adjusts based on screen size.

```
<div class="container">
  <div class="row">
    <div class="col-md-4">First column</div>
    <div class="col-md-4">Second column</div>
    <div class="col-md-4">Third column</div>
  </div>
</div>
```

6. Mention the uses of typography in Bootstrap.

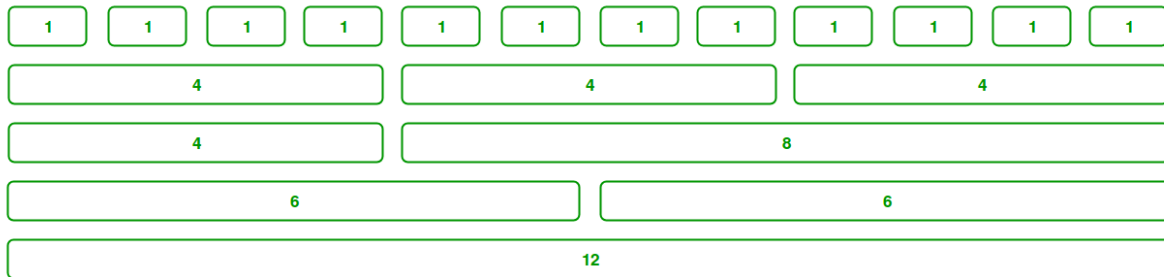
Typography can be used to create,

- Headings
- Subheadings
- Font color, font type, and alignment
- Lists
- Other inline elements

(PART B – 13 Marks)

1. Explain in detail about the bootstrap system which uses a series of containers, rows, and columns to layout and align content.

Bootstrap Grid System allows up to 12 columns across the page. You can use each of them individually or merge them together for wider columns. You can use all combinations of values summing up to 12. You can use 12 columns each of width 1, or use 4 columns each of width 3 or any other combination.



(i) Bootstrap Grid Classes

The Bootstrap grid system has five tiers of classes,

- col-xs for portrait phones (<576px)
- col-sm for landscape phones (>576px)
- col-md for tablets (>768px)
- col-lg for smaller desktops (>992px)
- col-xl for larger desktops (>1200px)

(ii) Bootstrap Grid Components

- **Containers:** Bootstrap requires a containing element to wrap site contents and house our grid system. Use 'container' for a responsive fixed width container and use 'container-fluid' for a full width container, spanning the entire width of your viewport.
- **Rows:** Rows are used to create a horizontal group of columns.
- **Columns:** Grid columns are created by specifying the number of twelve available columns.

(iii) Grid Rules

- Grid column elements are placed inside row elements, which create horizontal groups of columns.
- You can have as many rows as you want on the page, but columns must be immediate children of rows.
- In a full row, the column widths will be any combination that adds up to 12, but it is not mandatory to use all 12 available columns.

Example

```
<html>

  <head>

    <title> Sample Grid </title>

  </head>

  <body>

    <div class="container">

      <div class="row">

        <div class="col-md-6">First column</div>

        <div class="col-md-6">Second column</div>

      </div>

      <div class="row">

        <div class="col-md-4">First column</div>

        <div class="col-md-4">Second column</div>

        <div class="col-md-4">Third column</div>

      </div>

    </div>

  </body>

</html>
```

Sample Output

First Column	Second Column
--------------	---------------

First Column	Second Column	Third Column
--------------	---------------	--------------

2. Discuss how button classes can be used to create buttons of different sizes, styles, and behaviors that adapt to various screen sizes and devices. Include examples to illustrate how responsive design is achieved through Bootstrap's button utilities.

Bootstrap provides different classes that can be used with different tags, such as <button>, <a>, <input>, and <label> to apply custom button styles.

(i) Bootstrap Button Classes

S.No.	Class & Description
1	btn Default/ Standard button.
2	btn-primary Provides extra visual weight and identifies the primary action in a set of buttons.
3	btn-success Indicates a successful or positive action.
4	btn-info Contextual button for informational alert messages.
5	btn-warning Indicates caution should be taken with this action.
6	btn-danger Indicates a dangerous or potentially negative action.
7	btn-link Deemphasize a button by making it look like a link while maintaining button behavior.
8	btn-light Button with plain white background.
9	btn-dark Button with black background.

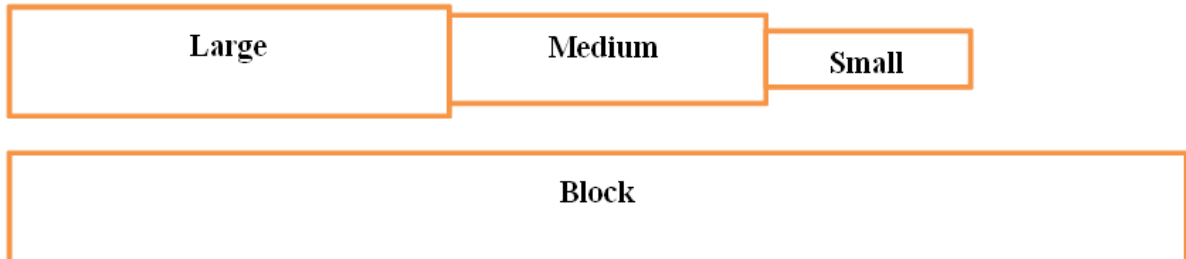
Example

```
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-secondary">Secondary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-light">Light</button>
```

(ii) Button Sizes

Bootstrap provides different classes that allow changing the size of the button.

- btn-lg → large buttons.
- btn-md → medium buttons.
- btn-sm → small buttons.
- btn-block → takes all width of parent element.



Example

```
<button type="button" class="btn btn-success btn-sm"> Small Button </button>
<button type="button" class="btn btn-success btn-md"> Medium Button </button>
<button type="button" class="btn btn-success btn-lg"> Large Button </button>
<button type="button" class="btn btn-success btn-block"> Block Button </button>
```

(iii) Button States

- active → The .active class is used to make button and link to **active state**.
- disabled → The disabled attribute is used with <button> element to set the **disabled state of the button**.

Example

```
<button type="button" class="btn btn-success active">Active Button</button>
<button type="button" class="btn btn-success" disabled>Disabled Button</button>
```

Benefits:

- **Consistency**
 - Buttons retain a **consistent appearance** across devices.
- **Customization**
 - Easy to **adapt for various use cases** using **utilities**.
- **Usability**
 - Ensures buttons are **accessible and user-friendly** on **touch** and **desktop devices**.

3. Explain Bootstrap alert components in detail. Discuss their types, usage, and customization with suitable examples.

Bootstrap Alerts are used to provide an easy way to create predefined alert messages. Alert adds a style to your messages to make it more appealing to the users.

(i) Alert Classes

Alert	Description
.alert-success	Positive action
.alert-info	Informative
.alert-warning	Warning
.alert-danger	Negative action
.alert-primary	Important action
.alert-secondary	Less important action
.alert-light	Light alert box
.alert-dark	Dark alert box

Example

```
<div class="alert alert-success"> Success! Color: Green </div>
<div class="alert alert-info"> Info! Color: Light Blue </div>
<div class="alert alert-warning"> Warning! Color: Yellow </div>
<div class="alert alert-danger"> Danger! Color: Red </div>
<div class="alert alert-primary"> Primary! Color: Blue </div>
<div class="alert alert-secondary"> Secondary! Color: Light Grey </div>
<div class="alert alert-dark"> Dark! Color: Dark Grey </div>
<div class="alert alert-light"> Light! Color: White </div>
```

Sample Output

Success! Color: Green
Info! Color: Light Blue
Warning! Color: Yellow
Danger! Color: Red
Primary! Color: Blue
Secondary! Color: Light Grey
Dark! Color: Dark Grey
Light! Color: White

(ii) Alert Link

The Bootstrap utility class alert-link can be used for any alert box links to instantly produce a matching colored links.

Example

```
<div class="alert alert-primary">  
  This is a primary alert with <a href="#" class="alert-link">an example link</a>.  
</div>  
  
<div class="alert alert-danger">  
  This is a danger alert with <a href="#" class="alert-link">an example link</a>.  
</div>
```

(iii) Closing Alerts

Data attributes provide a simple and easy way to add close functionality to the alert boxes. Just add the data-bs-dismiss="alert" to the close button and it will automatically enable the dismissal of the containing alert message box. Also, add the class .alert-dismissible to the .alert element for proper positioning of the .btn-close button.

Example

```
<div class="alert alert-success alert-dismissible">  
  <button type="button" class="close" data-dismiss="alert">×</button>  
  Success!  
</div>  
  
<div class="alert alert-info alert-dismissible">  
  <button type="button" class="close" data-dismiss="alert">×</button>  
  Info!  
</div>
```

(iv) Animated Alerts

The utility classes .fade and .show create the animation effect, when you close an alert message.

Example

```
<div class="alert alert-success alert-dismissible fade show">  
  <button type="button" class="close" data-dismiss="alert">×</button> Success!  
</div>
```

4. Explain in detail about changing the color of a text and its background in bootstrap.

A) Text Colors

Bootstrap provides many classes to set the text color of an element.

Example

```
<div class="container">
  <h2>Bootstrap Colors List</h2>
  <ul>
    <li class="text-muted">.text-muted class</li>
    <li class="text-primary">.text-primary class</li>
    <li class="text-success">.text-success class</li>
    <li class="text-info">.text-info class</li>
    <li class="text-warning">.text-warning class</li>
    <li class="text-danger">.text-danger class</li>
    <li class="text-secondary">.text-secondary class</li>
    <li class="text-white">.text-white class</li>
    <li class="text-dark">.text-dark class</li>
    <li class="text-body">.text-body class</li>
    <li class="text-light">.text-light class</li>
  </ul>
</div>
```

(i) Adding opacity to the text

There are two classes associated with text color in Bootstrap 5 that add opacity to the text.

- **text-black-50**
 - This represents a black colored text having 50% opacity with a white background.
- **text-white-50**
 - This represents a white colored text having 50% opacity with a black background.

Example

```
<p class="text-black-50">
  This paragraph has a black color, 50% opacity and white background</p>
<p class="text-white-50 bg-dark">
  This paragraph has a white color, 50% opacity and a black background</p>
```


B) Background Colors

Bootstrap provides many classes to set the background color of an element.

Example

```
<div class="container">
  <h2>Bootstrap Background Colors List</h2>
  <ul>
    <li class="bg-primary">.bg-primary class</li>
    <li class="bg-white">.bg-white class</li>
    <li class="bg-success">.bg-success class</li>
    <li class="bg-info">.bg-info class</li>
    <li class="bg-warning">.bg-warning class</li>
    <li class="bg-muted">.bg-muted class</li>
    <li class="bg-danger">.bg-danger class</li>
    <li class="bg-secondary">.bg-secondary class</li>
    <li class="bg-dark">.bg-dark class</li>
    <li class="bg-light">.bg-light class</li>
  </ul>
</div>
```

(i) Background Gradient

Bootstrap provides many classes to set the background gradient color of an element.

Example

```
<div class="container">
  <h2>Bootstrap Background Gradient Colors List</h2>
  <ul>
    <li class="bg-gradient bg-primary">.bg-primary gradient class</li>
    <li class="bg-gradient bg-white">.bg-white gradient class</li>
    <li class="bg-gradient bg--success">.bg-success gradient class</li>
    <li class="bg-gradient bg--info">.bg-info gradient class</li>
    <li class="bg-gradient bg--warning">.bg-warning gradient class</li>
    <li class="bg-gradient bg--muted">.bg-muted gradient class</li>
    <li class="bg-gradient bg--danger">.bg-danger gradient class</li>
    <li class="bg-gradient bg--secondary">.bg-secondary gradient</li>
    <li class="bg-gradient bg--dark">.bg-dark gradient class</li>
    <li class="bg-gradient bg--light">.bg-light gradient class</li>
  </ul>
</div>
```

5. Discuss different classes used for styling, responsiveness, and interactive features of tables in Bootstrap. Provide suitable examples to illustrate each type.

Bootstrap provides a series of classes that can be used to apply various styling to the tables such as changing the heading appearance, making the rows stripped, adding or removing borders, making rows hoverable, etc.

(i) Simple Table

The `.table` class is used to create a simple Bootstrap table.

Syntax

```
<table class="table"> Table Contents... </table>
```

(ii) Striped rows

The `.table-striped` class is used to create alternate dark and light rows.

Syntax

```
<table class="table table-striped"> Table Contents... </table>
```

(iii) Bordered Table

The `.table-bordered` class is used to add borders on all sides of the table and cell.

Syntax

```
<table class="table table-bordered"> Table Contents... </table>
```

(iv) Hover Rows Table

The `.table-hover` class is used to add the hover effect (change background color to gray when the mouse moves over) on table rows.

Syntax

```
<table class="table table-hover"> Table Contents... </table>
```

(v) Black/Dark Table

The `.table-dark` class is used to add the black background color of a table.

Syntax

```
<table class="table table-dark"> Table Contents... </table>
```

(vi) Dark Striped Table

The `.table-dark` and `.table-striped` classes are used to create a dark striped table.

Syntax

```
<table class="table table-dark table-striped"> Table Contents... </table>
```

(vii) Dark Hoverable Table

The `.table-dark` and `.table-hover` classes are used to add the hover effect (change background color to dark gray when the mouse moves over) on table rows.

Syntax

```
<table class="table table-dark table-hover"> Table Contents... </table>
```

(viii) Borderless Table

The `.table-borderless` is used to remove the border from the table.

Syntax

```
<table class="table table-borderless"> Table Contents... </table>
```

(ix) Colored Table

Bootstrap provides a number of contextual classes that can be used to color the entire row or a single cell of a table. The list of contextual classes is given below.

table-primary	table-secondary	table-success
table-danger	table-warning	table-info
table-light	table-dark	table-active

Example

```
<table class="table">
  <tr class="table-primary">
    <th>Reference No.</th>
    <th>Name</th>
  </tr>
  <tr class="table-secondary">
    <td>101</td>
    <td>ABC</td>
  </tr>
</table>
```

(x) Dark Colored Table

For creating dark colored tables, `.bg` conceptual classes are added to the `<tr>`.

Example

```
<table class="table table-dark">
  <tr class="bg-primary">
    <th>Reference No.</th>
    <th>Name</th>
  </tr>
  <tr class="bg-secondary">
    <td>101</td>
    <td>ABC</td>
  </tr>
</table>
```

6. Design a simple homepage layout for a pharmacy website named PharmEasy using HTML and Bootstrap. The page should include a header, basic navigation bar, and display at least three featured medical products with images, names, descriptions, and prices in rupees.

```

<html>
  <head>
    <title>PharmEasy - Your Trusted Pharmacy</title>
    <!-- Bootstrap CSS -->
  </head>
  <body>

    <!-- Header -->
    <header class="bg-success text-white text-center">
      <h1>PharmEasy</h1>
      <p>Your Trusted Online Pharmacy</p>
    </header>

    <!-- Navigation -->
    <nav class="navbar navbar-dark">
      <div class="container">
        <a class="navbar-brand" href="#">PharmEasy</a>
        <ul>
          <li><a href="home.html">Home</a></li>
          <li><a href="products.html">Products</a></li>
          <li><a href="offers.html">Offers</a></li>
          <li><a href="contact.html">ContactUs</a></li>
        </ul>
      </div>
    </nav>

    <!-- Featured Products -->
    <section class="container">
      <h2 class="text-center">Featured Products</h2>
      <div class="row">

        <!-- Product 1 -->
        <div class="col-md-3">
          <div class="card">
            
            <h5 class="card-title">Paracetamol</h5>
            <p>₹50</p>
          </div>
        </div>
      </div>
    </section>
  </body>
</html>

```

```
<!-- Product 2 -->
<div class="col-md-3">
  <div class="card">
    
    <h5 class="card-title">Prozac</h5>
    <p>₹60</p>
  </div>
</div>

<!-- Product 3 -->
<div class="col-md-3">
  <div class="card">
    
    <h5 class="card-title">Iron Red</h5>
    <p>₹150</p>
  </div>
</div>

<!-- Product 4 -->
<div class="col-md-3">
  <div class="card">
    
    <h5 class="card-title">Pyremol</h5>
    <p>₹250</p>
  </div>
</div>

</div>
</section>

<!-- Footer -->
<footer class="bg-dark text-white text-center">
  <p>© 2025 PharmEasy. All rights reserved.</p>
</footer>

<!-- Bootstrap JS -->

</body>
</html>
```

PART C – 15 Marks

1. Describe the Bootstrap component that improves user experience by providing intuitive and organized navigation across a website. How does it help users quickly access different sections and search for specific content with ease?

A navigation bar is used in every website to make it more user-friendly so that the navigation through the website becomes easy and the user can directly search for the topic.

(i) Basic navbar

It aligns the links on the left by default.

Example

```
<nav class="navbar navbar-default">
  <ul class="nav navbar-nav">
    <li><a href="#">Page 1</a></li>
    <li><a href="#">Page 2</a></li>
  </ul>
</nav>
```

(ii) Inverted navbar

It aligns the link on the left and inverts the color of the background.

Example

```
<nav class="navbar navbar-inverse">
  <ul class="nav navbar-nav">
    <li><a href="#">Page 1</a></li>
    <li><a href="#">Page 2</a></li>
  </ul>
</nav>
```

(iii) Colored navbar

To change the background colour of the navigation bar we can use various classes with the nav tag,

- bg-primary – Blue
- bg-success – Green
- bg-warning – Yellow
- bg-danger – Red
- bg-secondary – Grey
- bg-dark – Black
- bg-light – White

Example

```
<nav class="navbar navbar-default">
  <ul class="nav navbar-nav">
    <li class="bg-primary"><a href="#">Page 1</a></li>
    <li class="bg-success"><a href="#">Page 2</a></li>
    <li class="bg-danger"><a href="#">Page 3</a></li>
  </ul>
</nav>
```

(iv) Right-Aligned navbar

By default, the links on a navigation bar are arranged on the left. To align the links on the right side, navbar-right can be used.

Example

```
<nav class="navbar navbar-default">
  <ul class="nav navbar-nav">
    <li><a href="#">Page 1</a></li>
    <li><a href="#">Page 2</a></li>
  </ul>
  <ul class="nav navbar-nav navbar-right">
    <li><a href="#">Login</a></li>
    <li><a href="#">Sign Up</a></li>
  </ul>
</nav>
```

(v) Fixed navbar

We can fix a navigation bar at the top of the webpage so that, if you scroll down that webpage you can still view the navigation bar from any position on the page.

Example

```
<nav class="navbar navbar-inverse navbar-fixed">
  <ul class="nav navbar-nav">
    <li><a href="#">Page 1</a></li>
    <li><a href="#">Page 2</a></li>
    <li><a href="#">Page 3</a></li>
  </ul>
</nav>
```

(vi) Dropdown navbar

The benefit of this approach is that the whole navigation bar does not get occupied with these links.

Example

```
<nav class="navbar navbar-default">
  <ul class="nav navbar-nav">
    <li class="dropdown"><a href="#">Page 1</a></li>
    <ul class="dropdown-menu">
      <li><a href="#">Page 1-1</a></li>
      <li><a href="#">Page 1-2</a></li>
      <li><a href="#">Page 1-3</a></li>
    </ul>
    <li><a href="#">Page 2</a></li>
    <li><a href="#">Page 3</a></li>
  </ul>
</nav>
```

2. Develop a fully responsive website for “Delicious Bites Restaurant” using Bootstrap 5 that offers an interactive user experience and provides valuable, actionable feedback to enhance customer engagement.

```
<html>
<head>
  <title>Delicious Bites Restaurant</title>
</head>
<body>
  <!-- Navbar -->
  <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
    <div class="container">
      <a class="navbar-brand" href="#">Delicious Bites</a>
      <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs-target="#navbarNav">
        <span class="navbar-toggler-icon"></span>
      </button>
      <div class="collapse navbar-collapse" id="navbarNav">
        <ul class="navbar-nav ms-auto">
          <li class="nav-item"><a class="nav-link" href="#about">About</a></li>
          <li class="nav-item"><a class="nav-link" href="#menu">Menu</a></li>
          <li class="nav-item"><a class="nav-link"
href="#testimonials">Testimonials</a></li>
          <li class="nav-item"><a class="nav-link" href="#contact">Contact</a></li>
        </ul>
      </div>
    </div>
  </nav>

  <!-- Hero Section -->
  <header class="bg-primary text-white text-center py-5">
    <div class="container">
      <h1>Welcome to Delicious Bites</h1>
      <p class="lead">Experience the taste of perfection</p>
    </div>
  </header>

  <!-- About Section -->
  <section id="about" class="py-5">
    <div class="container text-center">
      <h2>About Us</h2>
      <p>Delicious Bites offers a variety of gourmet dishes crafted with passion and fresh
ingredients.</p>
    </div>
  </section>
```



```
<!-- Menu Section -->
<section id="menu" class="py-5 bg-light">
  <div class="container text-center">
    <div class="row">
      <div class="col-md-4">
        
        <h4>Idli</h4>
        <p>Steamed rice cakes served with chutney and sambar.</p>
      </div>
      <div class="col-md-4">
        
        <h4>Dosa</h4>
        <p>Thin, crispy crepe made from fermented rice and lentil batter.</p>
      </div>
      <div class="col-md-4">
        
        <h4>Aloo Paratha</h4>
        <p>Stuffed whole wheat bread served with yogurt and pickle.</p>
      </div>
    </div>
  </div>
</section>

<!-- Testimonials Section -->
<section id="testimonials" class="py-5">
  <div class="container text-center">
    <h2>Customer Feedback</h2>
    <blockquote class="blockquote">
      <p>"Absolutely loved the food! Highly recommended."</p>
      <footer class="blockquote-footer">Happy Customer</footer>
    </blockquote>
  </div>
</section>

<!-- Contact Section -->
<section id="contact" class="py-5 bg-light">
  <div class="container text-center">
    <h2>Contact Us</h2>
    <p>Email: contact@deliciousbites.com | Phone: (123) 456-7890</p>
  </div>
</section>

<!-- Footer -->
<footer class="bg-dark text-white text-center py-3">
  <p>&copy; 2025 Delicious Bites. All rights reserved.</p>
</footer>
</body> </html>
```